

Analysis of Abnormal Driver Behavior from Gaze Direction and Blinking

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Abstract. Most global road accidents are mainly caused by abnormal driving behaviors such as reckless driving, distraction, mobile phone usage, adjusting radios, or other vehicle systems. Driving while tired or drowsy can directly lead to accidents due to these abnormal driving behaviors. Additionally, this can result in property damage and pose risks to oneself and others, causing injuries or fatalities. Abnormal driving behaviors can be identified through facial expressions, yawning, blinking excessively, or looking elsewhere. Signs of fatigue include excessive blinking. To analyze factors contributing to road accidents, researchers have proposed the Unified Driver Attention and Fatigue Detection (UDAF) algorithm. This algorithm aims to enhance the efficiency of detecting abnormal driving behaviors by identifying gaze direction and blinking using Virtual Geometry Group (VGG16) and Long Short-Term Memory (LSTM). In this study, experiments were conducted using images and videos from front-facing cameras. The research achieved an average accuracy rate of 93.12% in detecting gaze direction and blinking through the UDAF algorithm.

Keywords: eye direction, eye blink, VGG16, LSTM

1 Introduction

Driving on the road is a complex activity that requires rapid decision-making. Drivers must maintain focus and remain aware of their surroundings at all times to prevent potential accidents. One of the most crucial driving behaviors is looking left and right, as it allows drivers to detect obstacles and other vehicles that may pose potential hazards. Additionally, driving while drowsy significantly impairs a driver's response ability, thereby increasing the risk of severe accidents. The behaviors of looking left and right, as well as drowsiness, are key factors that directly influence driving safety on the road. The detection and analysis of these behaviors can help identify issues related to

accident occurrence. Eye-tracking technology is one of the most effective methods for detecting these behaviors. Our research focuses on utilizing OpenCV and Mediapipe to detect and analyze driving behaviors based on eye movement. This approach employs Mediapipe Iris technology to enhance the accuracy of identifying key points in the eyes and analyzing left and right gaze direction, as well as blinking patterns, to compute relevant results. We aim to improve the efficiency of stress level analysis and assessment.

2 Related Works

A study [1] found that roadside advertisements negatively impact driver attention and road safety. Eye movement tracking (gaze) serves as a key variable in examining the relationship between driving characteristics, roadside environment conditions, and driver behavior. Autonomous driving enables drivers to focus on other activities [2], emphasizing the importance of monitoring driver attention to ensure a safe transition in control. The study suggests that head-up displays (HUDs) effectively assist drivers in maintaining focus on the road. The significance of accurately classifying eye movements [3], particularly during driving, was highlighted in another study. Other researchers [4], investigated the role of visual information on screens as a primary cause of driver distraction, employing attention maps and eye-tracking data. Their findings indicated that road paths in the upper and central areas of the driver's visual field attract more attention than roadside billboards. Pradeep Narayana and Nada Attar [5], identified distracted driving as a major cause of road accidents worldwide. Their study used a driving simulator to analyze eye movements and pupil size, which serve as indicators of cognitive load.

Their findings highlight the importance of driver response in SAE Level 3 and lower autonomous driving for safety. Delayed reactions were linked to reduced large eye movements and increased mid-sized movements [6]. Additionally, studies on blink detection and emotion recognition showed that driver emotions, including drowsiness, can be assessed through blink rate and eye movement patterns, with decreased blinking increasing accident risk [7]. This study explores eye movement analysis in driving safety. A MS-1D-CNN model [8] detects slow eye movements (SEMs) linked to driver sleep onset, achieving over 98% accuracy. Hazard prediction research [9] found that saccades, fixations, and blinks help assess risk, with a feedforward neural network predicting hazards at 70% accuracy.

For drowsiness detection [10], a VGG16-LGBM approach achieved 99% accuracy in 0.00829 seconds, using eye imagery instead of intrusive sensors. Additionally, HMI research [11] showed that visual interfaces improve situation awareness and eye-tracking behavior during automated driving transitions.

These findings highlight the importance of eye movement analysis in drowsiness detection, hazard prediction, and driver assistance, improving road safety and autonomous vehicle transitions. Research on human intent prediction [12] combined eye movement patterns with deep learning, achieving 97.42% accuracy, surpassing gaze-based methods. A fatigue detection system [13] using PERCLOS tracked eye aspect

ratio and issued alerts after 4 seconds of inactivity. Additionally, a blink detection algorithm [14] with an adaptive camera and k-means clustering improved real-time drowsiness monitoring. A computer vision-based system [15] further enhanced blink and facial movement tracking using a webcam. These advancements reinforce real-time driver monitoring, helping prevent fatigue-related accidents and ensuring safer driving experiences.

3 Methodology

Driving on the road requires a high level of caution and alertness. Observing the surrounding environment and responding quickly to changes are crucial for safe driving. Driver abnormalities, such as drowsiness, are among the leading causes of accidents, as they slow reaction times, impair decision-making, and may result in loss of vehicle control. Therefore, detecting and analyzing factors influencing driver behavior is essential for enhancing road safety. To achieve this objective, we conducted research to understand how to detect driving irregularities and assess driver attention.

The system these conditions consists of a workflow that integrates data processing, left and right gaze detection, and driver behavior analysis. The steps include image resizing and color adjustment, eye position detection, and blink analysis to identify driving abnormalities [16]. The detailed workflow for each step is illustrated in Fig. 1.

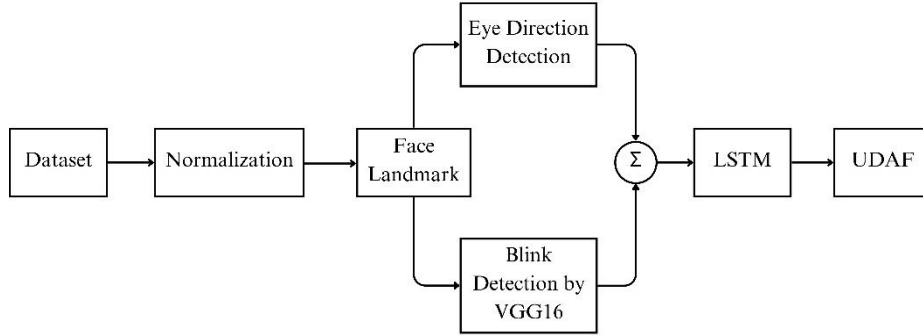


Fig. 1. System Workflow

1) The dataset is categorized based on the specific eye movement features to be detected. The prepared dataset is then standardized and visualized using histograms to enhance clarity and facilitate accurate detection. 2) Preprocessed and optimized images are used to identify key eye landmarks, enabling the determination of gaze direction in various orientations. 3) Video analysis is employed to track gaze direction and blinking patterns, ensuring data consistency. The collected data is then stored and refined to improve accuracy at each stage. 4) The UDAF (Unified Driver Attention Formula) is applied to assess overall driver alertness, providing an evaluation of the risk of drowsiness or loss of focus. This enhances the precision and comprehensiveness of abnormal behavior detection.

3.1 Data Size and Clarity Enhancement

The dataset used in this study focuses on gaze direction and blink detection, comprising 60 images depicting left, right, and forward gazes, along with 20 video clips demonstrating left and right gazes as well as blinking behavior. These data samples were sourced from Kaggle.com. Each image and video in the dataset were preprocessed using algorithms designed to enhance detection performance. The preprocessing techniques employed include normalization, which standardizes brightness levels, and histogram equalization, which improves contrast and highlights eye-related features essential for accurate detection. Involves selecting an appropriate dataset to test our detection system. To ensure comprehensive evaluation, we examine various eye movements, including looking forward, left, and right. Additionally, we verify specific conditions such as eye closure and yawning. Fatigue is also considered as a contributing factor affecting eye behavior, which is crucial for accurate detection and analysis.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (1)$$

Eq. (1) Min-Max Normalization. In the next step, the Min-Max scaling method is applied to normalize the data within the range [0,1]. This is done by subtracting the minimum value (Min) of a given feature/column and dividing by the range (Max – Min). Normalization is crucial because:

- Reduces feature imbalance: Features with higher values may have a greater impact than those with lower values if no adjustments are made. Normalization ensures balanced feature contributions.
- Enhances computational efficiency: Simplifies calculations and accelerates the learning process.
- Prevents overfitting: Helps models learn effectively from closely distributed data. Next, histogram equalization is applied to redistribute pixel intensity frequencies in an image. In a histogram, the x-axis represents pixel intensity values, while the y-axis shows their frequency of occurrence. This technique is particularly useful for grayscale images and enhances image details.
- Increases image contrast: Improves the visibility of finer details.
- Enhances feature detection: Particularly effective for improving low-contrast or underexposed images.

3.2 Detection

After adjusting image size, color consistency, and detection accuracy, we develop a script using OpenCV and Mediapipe, specifically leveraging the Iris module to detect eye characteristics. This script is designed to identify gaze direction (left, right) and count blink frequency, as illustrated in Fig. 2. To assess detection accuracy, 60 images are processed to evaluate gaze direction and blink count detection. Once satisfactory results are obtained from these images, the script is further tested using 10 video clips to evaluate its real-time performance in detecting gaze direction and counting blinks.

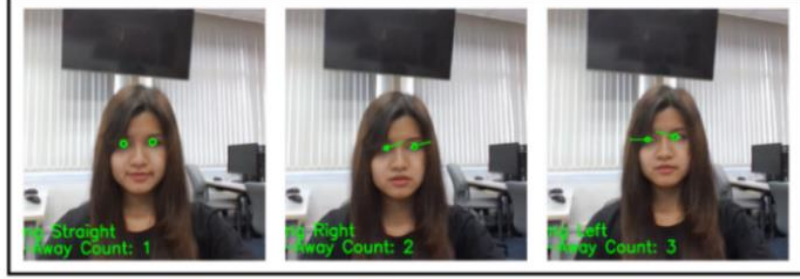


Fig. 2. Mediapipe Iris

During this phase, detection parameters are fine-tuned to maximize accuracy as needed. In the final step, the optimized script calculates the duration of the driver's gaze in each direction and the total number of blinks per clip. These results are then compared against ground truth data from the videos. Mathematical adjustments are applied to synchronize recorded time with actual video timestamps, ensuring that the duration measured by OpenCV accurately aligns with real-world time in the clips.

3.3 Result searching Process

Eye tracking is utilized to predict driving behavior, which is divided into two main aspects: counting the number of blinks to assess fatigue levels and measuring the duration of non-responsiveness to analyze left-right gaze behavior. Excessive or prolonged blinking can indicate drowsiness or fatigue. Previous research has shown that drowsy drivers tend to blink more frequently and for longer durations than usual. In this study, video recordings are used to track and count instances of complete eye closure to evaluate abnormality levels and calculate the blink rate. If the measured blink rate exceeds the predefined threshold, it suggests that the driver may exhibit abnormal behavior.

$$\theta x = \arctan \arctan \left(\frac{xg - xe}{zg - ze} \right) \quad (2)$$

$$\theta y = \arctan \arctan \left(\frac{yg - ye}{zg - ze} \right) \quad (3)$$

$$Drowsiness_{Level} = \frac{Total_Blink}{Total_Time} \quad (4)$$

Based on Equations 2 and 3, which pertain to direction detection and fatigue assessment through blinking, we employ the VGG16 model, pre-trained on the ImageNet dataset, to extract features from video frames of drivers, focusing on blink occurrences and gaze direction. After feature extraction, a Long Short-Term Memory (LSTM) network is utilized to predict the frequency and direction of blinking. Experimental results indicate that the combination of VGG16 and LSTM effectively detects blinking and gaze direction with high accuracy. Moreover, LSTM further facilitates the temporal analysis of behavioral changes in drivers.

3.4 Object Recognition

After detecting eye movement, the system's accuracy is evaluated using the **F1 score**, which balances **Precision** (correct detections out of all detected cases) and **Recall** (correct detections out of total actual cases). As the **harmonic mean** of these metrics, the **F1 score** provides a comprehensive measure of detection performance. Results and Discussion

3.5 Dataset and Evaluation Metrics

The dataset used in this study focuses on gaze direction and blink detection, comprising 60 images depicting left, right, and forward gazes, along with 20 video clips demonstrating left and right gazes as well as blinking behavior. These data samples were sourced from Kaggle.com. Each image and video in the dataset were preprocessed using algorithms designed to enhance detection performance. The preprocessing techniques employed include normalization, which standardizes brightness levels, and histogram equalization, which improves contrast and highlights eye-related features essential for accurate detection.

Table 1. Detection Results

Video	Gaze Left	Gaze Right	Gaze Forward	Blink Rate	Fatigue Level
1	16	18	26	27	Low
2	4	11	25	37	Moderate
3	10	2	41	37	Moderate
4	10	2	12	43	Moderate
5	2	1	25	45	Moderate
6	5	2	17	39	Moderate
7	2	1	21	45	Moderate
8	12	13	35	47	Moderate
9	1	7	22	54	High
10	3	4	28	38	Moderate
11	2	4	4	41	Moderate
12	1	1	9	42	Moderate
13	12	5	43	43	Moderate
14	4	3	19	41	Moderate
15	2	1	9	42	Moderate
16	4	6	16	45	Moderate
17	2	1	22	54	High
18	2	5	36	49	High
19	3	0	29	5	Low
20	2	7	12	54	High

3.6 Key Findings

An analysis of 60 images and 20 video clips revealed that alert drivers exhibit a balanced gaze distribution, spending an approximately equal amount of time looking left, right, and straight ahead. In contrast, drivers exhibiting abnormal behavior tend to fixate in a single direction for prolonged periods, with a decreased frequency of gaze shifts. This pattern indicates reduced situational awareness and an increased risk of missing critical signals. The blink analysis further showed that drivers with abnormal behavior blink more frequently and for longer durations, which correlates with fatigue symptoms. According to Table 1, alert drivers spent an average of 16 times looking left, 18 times looking right, and 26 times looking forward over a 60-second period. Conversely, drivers with abnormal behavior demonstrated prolonged fixation and increased blink frequency, suggesting impaired attentiveness.

In the initial stage, the researchers converted video frames into static images with a resolution of 224×224 pixels, which were then fed into the VGG16 model, pre-trained on the ImageNet dataset, to extract key features for blink detection. The extracted feature sequences were subsequently processed by a Long Short-Term Memory (LSTM) model, which analyzes video frame sequences over different time intervals. LSTM is well-suited for this task as it can effectively handle complex long-term dependencies, enabling more accurate predictions of blinking behavior and gaze direction.

The LSTM model learns from the features extracted by VGG16 to predict and detect blinking with high precision, while also assessing the driver's abnormality level based on blink count and frequency, aiding in the detection of fatigue indicators. The training results of the VGG16-LSTM model are summarized in Table 2, which compares the performance of blink detection with and without VGG16 and LSTM. The results demonstrate that the combined VGG16-LSTM model significantly outperforms other approaches in both detection accuracy and blink prediction.

Table 2 presents the F1 score for gaze direction detection alone, computed using Equation 4. The F1 score for gaze direction detection combined with blink detection, utilizing VGG16 and LSTM. The results indicate that the proposed model achieves high accuracy in detecting both blinking and gaze direction. Furthermore, the model effectively distinguishes between alert and impaired driving behavior, making it a valuable tool for developing alert systems to prevent accidents caused by driver fatigue.

$$F1\ Score = 2 \times \frac{Precision+Recall}{Precision \times Recall} \quad (5)$$

$$UDAF = 0.4 \cdot (EAR_{avg}) + 0.3 \cdot (Gazex + Gazey) + 0.2 \cdot Turncount + 0.1 \cdot Fatiguelevel \quad (6)$$

By integrating key features such as blink detection (Eye Aspect Ratio - EAR), gaze direction, head movement, and fatigue levels, the UDAF equation provides a comprehensive assessment of driver alertness. Each component serves a distinct role in enhancing the accuracy of detecting abnormal behaviors:

- EAR (Eye Aspect Ratio) is used to track blink frequency and duration, which are critical indicators of fatigue and visual impairment.

- Gaze direction monitors the driver's eye movements to analyze shifts in focus or signs of distraction.
- Turn count records head movements, which may indicate inattentiveness or fatigue.
- Fatigue level aggregates all relevant data to assess the risk of eye closure or loss of focus.

Table 2. Eye Direction and Blink Detection

Video	F1 Gaze Direction	F1 UDAF Equation
1	66.67%	91.50%
2	77.44%	95.77%
3	68.97%	92.47%
4	66.67%	94.43%
5	72.22%	93.24%
6	64.71%	93.30%
7	62.50%	91.22%
8	77.78%	95.21%
9	74.42%	94.83%
10	68.75%	91.88%
11	66.67%	91.00%
12	50.00%	91.57%
13	68.97%	92.71%
14	78.26%	94.57%
15	60.00%	93.79%
16	68.75%	92.50%
17	64.00%	92.81%
18	79.41%	93.81%
19	66.67%	93.62%
20	71.43%	92.23%

These features are incorporated into the UDAF equation, they provide a holistic perspective on driver conditions, significantly improving the early detection of fatigue and attention loss. Such detection plays a crucial role in preventing accidents, particularly in Advanced Driver Assistance Systems (ADAS) designed to enhance road safety. Furthermore, when the UDAF equation is integrated with the VGG16 and LSTM models, the system achieves even greater efficiency, as demonstrated in Table 2.

- VGG16 excels in extracting fine-grained spatial features from video frames, such as eye and facial movements.
- LSTM (Long Short-Term Memory) is well-suited for temporal sequence analysis, enabling efficient tracking of eye movement sequences, head turns, and behavioral changes over time.

The integration of these techniques enhances the accuracy and reliability of detecting driver abnormalities. The UDAF equation, when combined with VGG16 and LSTM, serves as a crucial tool in the development of ADAS, focusing on road safety enhancement.

4 Conclusion

This study focuses on analyzing abnormal driving behavior by monitoring gaze direction and blink rate to assess driving irregularities caused by inattention and fatigue. We designed a detection system using OpenCV and Mediapipe to track eye movements and measure blink frequency based on image and video data. Fig.3 illustrates the results, where the X-axis represents time in seconds within the video clip, while the Y-axis denotes the cumulative number of blinks recorded.

Red Line: The red line in the graph represents the detection of blinks throughout the video duration, indicating the blinks occurring at specific times.

Purple Box: The purple box in the graph highlights the onset of abnormal driving behavior, which may be caused by drowsiness, inattention, or fatigue. This is evaluated by detecting 20 or more blinks within a 60-second period, which may signal a loss of focus or driver fatigue. The training process begins with using the VGG16 model, pre-trained on a prior dataset to extract features from video frames. The LSTM network is then employed to classify the blink sequences and gaze direction. The results show that the trained model can accurately detect blinks and estimate gaze direction with an average accuracy rate of 93.12%, demonstrating its potential for real-world applications. In conclusion, this study confirms the feasibility and effectiveness of using gaze direction and blink detection as indicators of abnormal driver behavior.

In future research, the system could be enhanced to detect abnormal driver behavior under varied conditions such as lighting, head position, and eyewear. Although this aspect has not yet been tested, future studies will focus on improving model accuracy and utilizing advanced deep learning techniques to enhance system reliability.

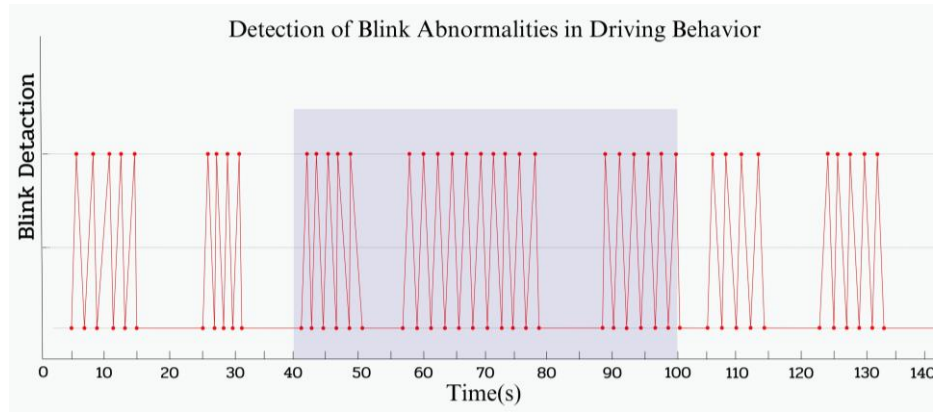


Fig. 3. Graph Showing Time Periods of Abnormal Behavior

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